

Survey participants

- Anand T., managing director, KNEWRON Technologies Pvt Ltd
- Bhaskar Rao, chief executive officer, Technosphere Labs Pvt Ltd
- Chandrashekhar Gampawar, ESDM consultant, Lets Win Globe
- Dileep Jain, sales head, Rajguru Electronics
- Hemant Karnat, chief technology officer, Shalaka Technologies Pvt Ltd
- Kautubha Parkhi, principal analyst, Insight Research
- Rajesh Sola, core faculty, C-DAC-ACTS
- Srinivasa Moorthy, chief executive officer, Andhra Pradesh Electronics & IT Agency
- Suniel Kumar G., director, Nexiot
- Yogesh Borthade, director and chief technology officer, YUPS Tech Solutions

plication. So the sheer volume of deployment will generate a huge demand for communication hardware, devices and also services. And this is where electronics plays a big role.

Each IoT device will require, at the minimum, a microcontroller to add intelligence to the device, one or more sensors to allow for data collection, one or more chips for connectivity and data transmission, and a memory component. Connected devices that transmit information across relevant networks rely on innovations from semiconductor players—highly-integrated microchip designs, for instance, and very low-power functions in certain applications.

Semiconductor companies that can effectively deliver these and other innovations to original equipment manufacturers, original device manufacturers and others that are building IoT products and applications will play an important role in the development of the market. That market, in turn, may turn out to be a significant growth opportunity for semiconductor players and their customers.

Technology trends

Wide adoption of the IoT may need customised technology solutions that are suitable for the country. According to the survey participants, low-power wide area network (LPWAN) technologies like LoRa are an excellent fit in the Indian context of smart cities and smart agricul-

ture. Sigfox, NB-IoT and the like are also gaining importance for the same reasons.

A new class of system(s)-on-chip (SoC) based devices, with optimal power and connectivity features as well as with sensor integration, will be in demand. First generation of such chips is already on its way, although it will probably be a few generations before the chips can deliver all the required functionality. Energy-harvesting, wireless integrated circuits can enable self-powered sensor nodes, which are easy to deploy without the need for power or signal wiring.

Additionally, a new array of sensors based on micro electro-mechanical systems technology is rapidly being developed to enable IoT applications beyond motion and image sensing, to include those that measure humidity, altitude, food calorie composition and various human health indicators. This is a result of the increasing variety of consumer IoT applications such as wearable and clustered systems.

With connectivity being a key capability required in IoT applications, multiple communication standards and protocols are being supported by consumer electronics and industrial devices, like low power and personal area wireless mesh networks, which are primarily suited for sensors.

IoT applications usually require a single chip in an acceptable form factor and low power consumption

for battery-operated devices. This requirement is a factor for growth in SoCs with embedded cores, GPUs and integrated wireless connectivity in a single package. Progress in these technologies has allowed the development of such IoT devices as sensors that have compute, storage and network capabilities built into extremely-small form factors, yet with low energy requirements.

On the services front, one emerging requirement will be asynchronous data transmission by the device. This will lower the cost of the device and the entire infrastructure. This will also need security in order to prevent tampering with the data transmitted by the device. And a reduction in ISO-OSI stack will help—working closer to the PHY layer will be better suited in terms of cost, power, carbon dioxide emissions, etc.

Moving forward

Like any other innovative concept, growth of the IoT also depends on several factors. We asked the survey participants to suggest the possible challenges that could derail the growth of the IoT sector in the country. Here is a collation of their opinions:

- Lack of awareness and limited availability of real products and services
- High cost of connectivity
- Inadequate security and privacy protection for user data
- Lack of consistent standards
- Technological limitations that affect functionality
- Discontinuous and fragmented IoT value chain
- Lack of cost-effective solutions
- Extended decision making cycles
- Lack of standardisation
- Regulatory overheads

However, the hope is that these problems are not insurmountable, particularly if the electronics industry players are willing to take an active role in solving these. **EFY**